

SingleRAN

LTE and NR Zero Bufferzone Feature Parameter Description

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Contents

1 Change History.....**1**

2 About This Document.....**3**

2.1 General Statements..... 3

2.2 Applicable RAT..... 3

2.3 Features in This Document..... 4

2.4 Differences..... 4

3 Overview.....**6**

4 LTE and NR Zero Bufferzone (TDD).....**8**

4.1 Principles..... 8

4.1.1 Interference Avoidance During PUSCH/PDSCH Scheduling..... 9

4.1.2 Interference Avoidance During Initial Access Scheduling..... 11

4.1.3 Bandwidth Adaptation for Interference Avoidance..... 11

4.1.4 Interference Avoidance During PUCCH Scheduling..... 12

4.1.5 PUCCH Frequency-Domain Position Optimization..... 13

4.1.6 Channel Adaptation Optimization..... 14

4.1.7 SRS SINR Correction..... 14

4.2 Network Analysis..... 15

4.2.1 Benefits..... 15

4.2.2 Impacts..... 15

4.3 Requirements..... 17

4.3.1 Licenses..... 17

4.3.2 Functions..... 17

4.3.2.1 Prerequisite Functions..... 17

4.3.2.2 Mutually Exclusive Functions..... 19

4.3.2.3 Function Impacts..... 20

4.3.3 Hardware..... 23

4.4 Operation and Maintenance..... 23

4.4.1 Data Configuration..... 23

4.4.1.1 Data Preparation..... 23

4.4.1.2 Using MML Commands..... 27

4.4.1.3 Using the MAE-Deployment..... 28

4.4.2 Activation Verification..... 28

4.4.3 Network Monitoring..... 29

5 LTE and NR Zero Bufferzone (FDD)..... 30

5.1 Principles..... 30

5.2 Network Analysis..... 33

5.2.1 Benefits..... 33

5.2.2 Impacts..... 33

5.3 Requirements..... 35

5.3.1 Licenses..... 35

5.3.2 Functions..... 35

5.3.2.1 Prerequisite Functions..... 36

5.3.2.2 Mutually Exclusive Functions..... 37

5.3.2.3 Function Impacts..... 40

5.3.3 Hardware..... 57

5.3.4 Others..... 58

5.4 Operation and Maintenance..... 58

5.4.1 Data Configuration..... 58

5.4.1.1 Data Preparation..... 58

5.4.1.2 Using MML Commands..... 60

5.4.1.3 Using the MAE-Deployment..... 61

5.4.2 Activation Verification..... 61

5.4.3 Network Monitoring..... 61

6 Parameters..... 62

7 Counters..... 64

8 Glossary..... 65

9 Reference Documents..... 66

1 Change History

The following describes the specific changes in different issues.

SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 01 (2025-03-10).

- Technical changes

Change Description	Parameter Change	Base Station Model
Added incompatibility of BTS5929R with the following functions: (1) LTE and NR zero bufferzone (TDD); (2) LTE and NR zero bufferzone (FDD). For details, see 4.3.3 Hardware and 5.3.3 Hardware .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

Change Description	Parameter Change	Base Station Model
Added support for LTE and NR zero bufferzone (FDD). For details, see 5 LTE and NR Zero Bufferzone (FDD) .	Added parameters: • CellAlgoExtSwitch.LnrZeroBufferzoneSw • CellAlgoExtSwitch.LnrCrslIntrfReduceStartTime • CellAlgoExtSwitch.LnrCrslIntrfReduceStopTime • CellAlgoExtSwitch.LteCrsPwrReduceAmount • CellAlgoExtSwitch.MaxLteCrsPwrReduceDuration • NRDUCellResShrCfg.LnrZeroBufferzoneSw • NRDUCellResShrCfg.LnrCrslIntrfReduceStartTime • NRDUCellResShrCfg.LnrCrslIntrfReduceStopTime • NRCeEutraNRelat.LteNrIntrflndicator	• 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R • DBS3900 LampSite and DBS5900 LampSite
Added an impact relationship between LNR CRS muting and RF module fast deep dormancy. For details, see 5.3.2.3 Function Impacts .	None	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

- Editorial changes
Revised descriptions in this document.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to LTE TDD, LTE FDD, and New Radio (NR).

2.3 Features in This Document

This document describes the following features.

RAT	Feature ID	Feature Name	Chapter/Section
LTE TDD	TDLBFD-191101	0 BufferZone	4 LTE and NR Zero Bufferzone (TDD)
LTE FDD	LCOFD-220302	LTE and NR Zero Bufferzone (LTE FDD)	5 LTE and NR Zero Bufferzone (FDD)
NR FDD	FOFD-100251	LTE and NR Zero Bufferzone (NR FDD)	

2.4 Differences

Differences between LTE FDD and LTE TDD

Function Name	Difference	Chapter/Section
LTE and NR zero bufferzone (TDD)	Supported only in LTE TDD	4 LTE and NR Zero Bufferzone (TDD)
LTE and NR zero bufferzone (FDD)	Supported only in LTE FDD	5 LTE and NR Zero Bufferzone (FDD)

Differences between NR FDD and NR TDD

Function Name	Difference	Chapter/Section
LTE and NR zero bufferzone (TDD)	Supported only in NR TDD	4 LTE and NR Zero Bufferzone (TDD)
LTE and NR zero bufferzone (FDD)	Supported only in NR FDD	5 LTE and NR Zero Bufferzone (FDD)

Differences between NSA and SA

Function Name	Difference	Chapter/Section
LTE and NR zero bufferzone (TDD)	None	4 LTE and NR Zero Bufferzone (TDD)
LTE and NR zero bufferzone (FDD)	None	5 LTE and NR Zero Bufferzone (FDD)

Differences between high frequency bands and low frequency bands

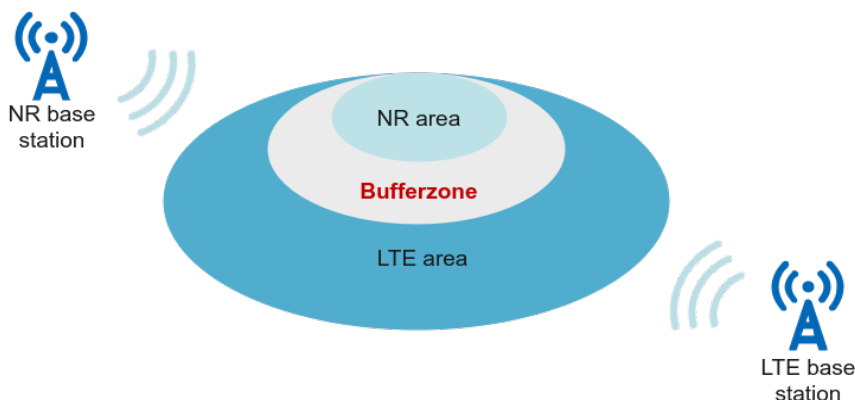
Function Name	Difference	Chapter/Section
LTE and NR zero bufferzone (TDD)	Supported only in low frequency bands	4 LTE and NR Zero Bufferzone (TDD)
LTE and NR zero bufferzone (FDD)	Supported only in low frequency bands	5 LTE and NR Zero Bufferzone (FDD)

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

3 Overview

LTE spectrum refarming takes time during the LTE-to-NR evolution, and not all LTE spectrum will be refarmed. As such, LTE and NR spectrum may overlap, causing NR cells to be interfered by intra-frequency neighboring LTE cells and affecting the NR cell throughput and user-perceived rate. To prevent the impact of neighboring LTE cells on NR cells, a physical bufferzone is required between the LTE and NR areas. LTE cells in the bufferzone cannot use the overlapping LTE and NR spectrum resources as shown in [Figure 3-1](#).

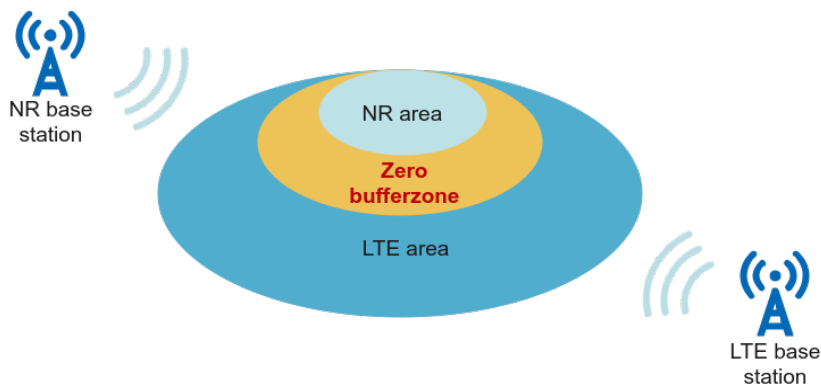
Figure 3-1 Common LTE and NR deployment



The bufferzone ensures the performance of the NR area. However, the intra-frequency LTE spectrum in the bufferzone cannot be used, resulting in the waste of LTE spectrum resources.

To alleviate LTE spectrum resource waste, LTE and NR zero bufferzone is introduced to activate some LTE spectrum resources in the bufferzone while exerting little impact on the performance of the NR area, as shown in [Figure 3-2](#).

Figure 3-2 Deployment of LTE and NR zero bufferzone

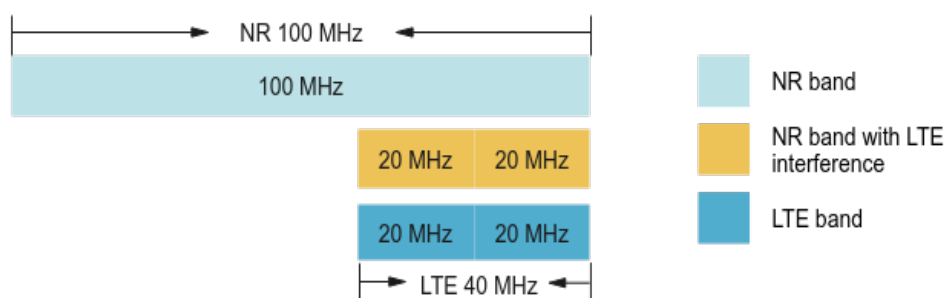


4 LTE and NR Zero Bufferzone (TDD)

4.1 Principles

As LTE spectrum refarming is time-consuming and not all LTE spectrum will be refarmed to carry NR services, some LTE bands still overlap with NR bands, resulting in intra-frequency interference between LTE and NR. Intra-frequency interference affects NR cell performance indicators, such as the throughput and user-perceived rate. This is particularly evident in the initial phase of NR network construction. Intra-frequency interference from LTE to NR may also occur under other combinations of frequency bands and cell bandwidths. In an example shown in [Figure 4-1](#), NR occupies 100 MHz of the 100 MHz contiguous spectrum, and LTE occupies 40 MHz. There is an overlapping 40 MHz bandwidth.

Figure 4-1 Overlapping between LTE and NR bands in the case of a 100 MHz NR cell



This feature enables the base station to take varying actions depending on the intra-frequency interference that neighboring LTE cells cause to NR UEs as follows:

- If the UEs experience strong LTE interference, the base station does not schedule them on the bands with LTE interference.
- If the UEs experience weak LTE interference, the base station does not avoid the bands with LTE interference during scheduling.

LTE and NR zero bufferzone includes the following functions:

- Interference avoidance during PUSCH/PDSCH scheduling

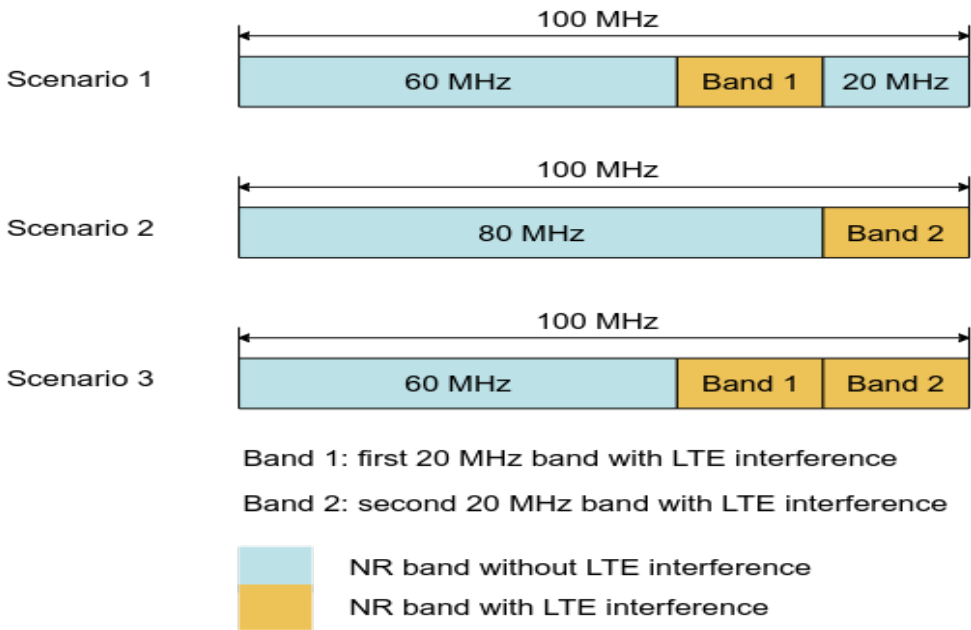
- Interference avoidance during initial access scheduling
- Bandwidth adaptation for interference avoidance
- Interference avoidance during PUCCH scheduling
- PUCCH frequency-domain position optimization
- Channel adaptation optimization
- SRS SINR correction

4.1.1 Interference Avoidance During PUSCH/PDSCH Scheduling

Figure 4-2 shows the band positions in three scenarios with LTE and NR intra-frequency interference.

In scenario 1, the interfered band is in the middle of the NR band. In scenario 2, the interfered band is located at one end of the NR band. In scenario 3, two interfered bands are located at one end of the NR band.

Figure 4-2 Band positions in a 100 MHz NR cell receiving intra-frequency interference from LTE



For NR UEs that would otherwise experience strong interference from neighboring LTE cells, the interference avoidance during PUSCH/PDSCH scheduling function is supported. To enable this function, the following parameters must be set:

- The **UL_CELL_LEVEL_INTRF_AVOID** and **DL_UE_LEVEL_INTRF_AVOID** options of the **NRDUCellIntrfIdent.IntrfOptimizationMethod** parameter are selected.
- The **NRDUCellIntrfIdent.IntrfIdentificationMethod** parameter is set to **UL_DL_MANUAL**.

After this function is enabled in an NR cell, cell reestablishment will be triggered. After the reestablishment, the gNodeB measures the uplink and

downlink channel quality in the interfered bands and non-interfered bands, and dynamically schedules bandwidth resources based on the channel quality, reducing the impact of LTE intra-frequency interference on NR.

This function is implemented as follows:

1. Interfered band configuration

Perform the following configurations after the bands where the cell is interfered by LTE are determined:

- Configure the indexes of the interfered bands using the **NRDUCellIntrfBand.InterferenceBandIndex** parameter. Up to two interfered bands can be configured.
- Configure the interfered bands. Specify the start and end RB positions using the **NRDUCellIntrfBand.StartRb** and **NRDUCellIntrfBand.EndRb** parameters, respectively, for each interfered band.
- Configure the interference direction by setting the **NRDUCellIntrfBand.InterferenceDirection** parameter to **UPLINK_DOWNLINK**.

For example, set the **NRDUCellIntrfBand.InterferenceBandIndex** parameter to **0** and specify the corresponding **NRDUCellIntrfBand.StartRb** and **NRDUCellIntrfBand.EndRb** parameters to determine the start and end RB positions of the first band with LTE interference (that is, Band 1 in [Figure 4-2](#)).

NOTICE

Some UEs may not support measurement of multiple interfered bands (shown in scenario 3), resulting in compatibility issues such as service drops. If UEs do not support measurement of multiple interfered bands and interference exists across the 40 MHz overlapping band, only one band (Band 1) with a bandwidth of 40 MHz needs to be set for interference measurement.

2. Avoidance during uplink scheduling

The gNodeB calculates the difference between the uplink interference strength in the interfered and non-interfered bands every 100 ms and compares the difference with the value of the **NRDUCellIntrfIdent.UlIntrfDiffThld** parameter.

- If the difference is greater than the value of this parameter, uplink scheduling cannot be performed in the interfered band.
- If the difference is less than or equal to the value of this parameter, uplink scheduling can be performed in the interfered frequency band.

3. Avoidance during downlink scheduling

The gNodeB performs downlink scheduling on the interfered frequency band only when all the following conditions are met:

- The uplink SRS interference strength difference between the interfered and non-interfered bands is less than or equal to the value of the **NRDUCellIntrfIdent.UlSrsIntrfDiffThld** parameter. This condition applies only when the **BW_EVAL_MCS_SEL_OPT_SW** option of the **NRDUCellIntrfIdent.ChnMeasCtrlSw** parameter is selected.

- The gNodeB determines that the spectral efficiency is higher when downlink scheduling is also performed on the interfered band. The spectral efficiency is estimated before downlink scheduling based on the channel quality and the spectral efficiency weighting coefficient for each range. In this case, the scheduling can be performed on the entire bandwidth of the NR cell.

The weighting coefficients are configured using the following parameters. A larger value indicates a higher weight for the corresponding band.

- The weighting coefficient for the spectral efficiency of the non-interfered band is determined by the **NRDUCellIntrfIdent.NolntrfBandSpectEffWtFactor** parameter.
- The weighting coefficient for the spectral efficiency of the non-interfered band and the first band with LTE interference is determined by the **NRDUCellIntrfIdent.IntrfBand1SpectEffWtFactor** parameter.
- The weighting coefficient for the spectral efficiency of the non-interfered band and the second band with LTE interference is determined by the **NRDUCellIntrfIdent.IntrfBand2SpectEffWtFactor** parameter.

4.1.2 Interference Avoidance During Initial Access Scheduling

In the initial access phase, the base station performs scheduling on the entire bandwidth (including both interfered and non-interfered bands) by default to obtain higher bandwidth gains. A cell will be interfered by a heavily loaded neighboring cell. As a result, a large number of block errors will occur during a UE's initial access to this cell, causing retransmissions. In this case, the retransmissions for this UE occupy RB resources of other UEs, affecting the overall user-perceived rate that can be achieved in the cell.

If the interference avoidance during initial access scheduling function is enabled, the base station preferentially schedules UEs in non-interfered bands during initial access.

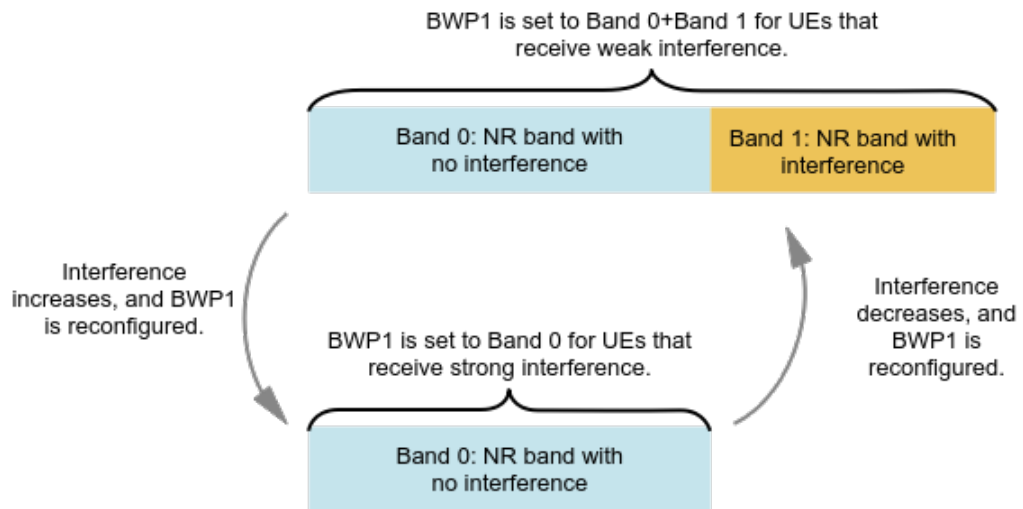
The **NONINTRF_SPCT_PRI_SW** option of the **NRDUCellIntrfIdent.ChnMeasCtrlSw** parameter can be selected to enable this function. For details about how to configure interfered bands, see [4.1.1 Interference Avoidance During PUSCH/PDSCH Scheduling](#).

4.1.3 Bandwidth Adaptation for Interference Avoidance

The bandwidth adaptation for interference avoidance function reconfigures the BWP1 for UEs to achieve interference avoidance, reducing the impact of LTE intra-frequency interference on NR.

This function requires that the interference avoidance during PUSCH/PDSCH scheduling function be enabled. If only the interference avoidance during PUSCH/PDSCH scheduling function is enabled, gNodeB achieves interference avoidance by scheduling UEs on the non-interfered bands. If the bandwidth adaptation for interference avoidance function is also enabled, interference avoidance is achieved by reconfiguring the BWP1 for UEs, as shown in [Figure 4-3](#).

Figure 4-3 Bandwidth adaptation for interference avoidance



BWP1 reconfiguration is implemented as follows:

- When the gNodeB determines that an interfered UE meets any of the following conditions, it reconfigures BWP1 of the UE to include only the non-interfered band. That is, BWP1 is changed from Band 0+Band 1 to Band 0.
 - The SRS interference difference between the interfered band and the non-interfered band is greater than the uplink SRS interference difference threshold. (That is, $\text{NR Band 1 SRS NI} - \text{NR Band 0 SRS NI} > \text{NRDUCellIntrfIdent.UlSrsIntrfDiffThld.}$)
 - The gNodeB determines that the performance of Band 0+Band 1 is inferior to that of Band 0 according to the estimation result.
- The gNodeB changes BWP1 from Band 0 to Band 0+Band 1 for a UE when identifying that the UE meets all of the following conditions:
 - The UE performs large-packet services.
 - BWP1 is changed from Band 0 to Band 0+Band 1 for a maximum of two times for the UE.
 - More than 10s has elapsed since the last time BWP1 is changed from Band 0+Band 1 to Band 0.
 - The path loss at the current time is 10 dB less than that at the previous BWP1 reconfiguration when BWP1 is changed from Band 0+Band 1 to Band 0.

The **INTRF_AVOID_BW_ADAPT_SW** option of the **NRDUCellIntrfIdent.ChnMeasCtrlSw** parameter can be selected to enable this function.

4.1.4 Interference Avoidance During PUCCH Scheduling

The interference avoidance during PUCCH scheduling function prevents the PUCCH from being transmitted on the interfered bands, reducing interference to the PUCCH and improving cell performance.

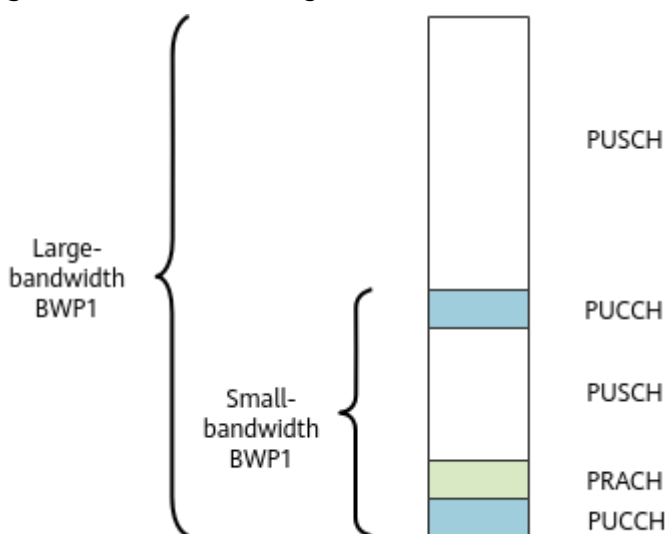
The **PUCCH_INTRF_AVOID_SW** option of the **NRDUCellAlgoSwitch.CommChnIntrfAvoidSwitch** parameter can be selected to

enable this function. However, the PUSCH's frequency-domain positions may be discontinuous. If there are UEs that do not support discontinuous PUSCH scheduling on the network, the uplink peak rates of these UEs may decrease.

4.1.5 PUCCH Frequency-Domain Position Optimization

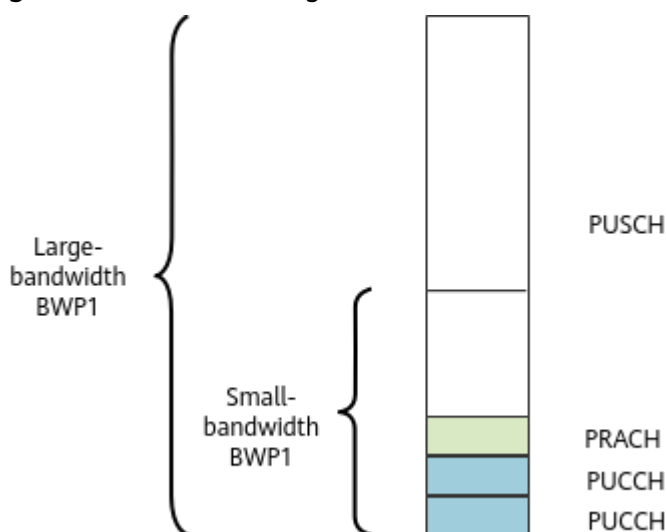
After the bandwidth adaptation for interference avoidance function enables the bandwidth of BWP1 to decrease for a UE, the PUCCH will be placed at both ends of the small-bandwidth BWP1. This separates the PUSCH in the original large-bandwidth BWP1, as shown in [Figure 4-4](#). If the UE does not support discontinuous PUSCH scheduling, the UE's performance may be affected.

Figure 4-4 PUCCHs configured at both ends of the small-bandwidth BWP1



To prevent the separation of the PUSCH, the PUCCH frequency-domain position optimization function is introduced. This function enables the PUCCH to be placed at one end of the small-bandwidth BWP1 (as shown in [Figure 4-5](#)), which enables that the PUSCH can be continuously scheduled.

Figure 4-5 PUCCHs configured at one end of small-bandwidth BWP1



The **PUCCH_FREQ_POS_OPT_SW** option of the **NRDUCellPucch.PucchAlgoSwitch** parameter can be selected to enable this function.

4.1.6 Channel Adaptation Optimization

The channel adaptation optimization function includes the following three sub-functions:

- Optimized downlink interference randomization-based scheduling
 When this function is enabled, non-interfered bands are configured as the start frequency-domain positions for scheduling in cells with different PCIs. For details, see *Downlink Scheduling in 5G RAN Feature Documentation*.
- Optimized SRS RSRP-based beam selection
 When this function is enabled, the optimal beam is selected based on the RSRP of SRS in the non-interfered bands instead of the full-bandwidth SRS RSRP. This function avoids the impact of the RSRP of SRS in the interfered bands.
- Optimized PDCCH frequency-domain range configuration
 When this function is enabled, non-interfered bands are configured as the frequency-domain position of CORESET1. This function prevents the decrease in the number of available CCE resources and mitigates CCE resource loss. CORESET1 span only in a contiguous frequency-domain range. If the non-interfered bands are discontinuous, the PDCCH capacity will decrease.

The **CHANNEL_ADAPT_OPT_SW** option of the **NRDUCellIntrfIdent.ChnMeasCtrlSw** parameter can be selected to enable this function.

4.1.7 SRS SINR Correction

In scenarios with strong interference, the measured SRS SINR may be higher than the actual value, which can cause the following problems:

- In the frequency domain, an interfered band is selected for SRS SINR-based scheduling.
- The base station selects an MCS index higher than required when the MCS index is selected based on the SRS SINR, increasing the block errors.

After the SRS SINR correction function is enabled, the gNodeB adjusts the SRS SINR to a value closer to the actual value.

The **UL_SINR_CORRECTION_SW** option of the **NRDUCellIntrfIdent.ChnMeasCtrlSw** parameter can be selected to enable this function.

4.2 Network Analysis

4.2.1 Benefits

This feature applies to the scenarios where a small number of LTE bands yet to be refarmed in NR coverage areas overlap the NR bands and impose intra-frequency interference on NR, causing NR network performance deterioration.

This feature increases the downlink throughput of NR UEs. The feature gains increase with the range and strength of the interference caused by the LTE on NR.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

After this feature is enabled, extra aperiodic CSI-RSs are reported in the interfered bands, resulting in lower CQIs in these bands. According to the existing counter measurement mechanism, the average wideband CQI in single-codeword transmission in a cell decreases.

Average wideband CQI in single-codeword transmission in a cell = $[(A_0 \times 0) + (A_1 \times 1) + (A_2 \times 2) + \dots + (A_{15} \times 15)] / (A_0 + A_1 + A_2 + \dots + A_{13} + A_{14} + A_{15})$

- A_0 : **N.ChMeas.CQI.SingleCW.0**
- A_1 : **N.ChMeas.CQI.SingleCW.1**
- A_2 : **N.ChMeas.CQI.SingleCW.2**
- A_3 : **N.ChMeas.CQI.SingleCW.3**
- A_4 : **N.ChMeas.CQI.SingleCW.4**
- A_5 : **N.ChMeas.CQI.SingleCW.5**
- A_6 : **N.ChMeas.CQI.SingleCW.6**
- A_7 : **N.ChMeas.CQI.SingleCW.7**
- A_8 : **N.ChMeas.CQI.SingleCW.8**
- A_9 : **N.ChMeas.CQI.SingleCW.9**
- A_{10} : **N.ChMeas.CQI.SingleCW.10**

- A_{11} : **N.ChMeas.CQI.SingleCW.11**
- A_{12} : **N.ChMeas.CQI.SingleCW.12**
- A_{13} : **N.ChMeas.CQI.SingleCW.13**
- A_{14} : **N.ChMeas.CQI.SingleCW.14**
- A_{15} : **N.ChMeas.CQI.SingleCW.15**
- After this feature is enabled, the base station delivers aperiodic CSI-RS measurement configurations to all UEs. As aperiodic CSI-RSs are reported on the PUSCH, the MCS index on the PUSCH decreases, the number of scheduling times and scheduling duration increase, and the average uplink UE throughput decreases. The throughput decrease is more obvious when the cell load is heavy.
- Enabling this feature will cause the cell to be automatically reestablished, causing service drops and re-access of admitted UEs. Therefore, you are advised to enable this feature when there are a small number of UEs in the cell.
- When the base station performs interference measurement, the system overheads increase. In addition, when the base station performs adaptive dynamic scheduling to avoid LTE interference, frequency-domain resources in the interfered bands cannot be utilized. The preceding impacts decrease system capacity to some extent. Slight performance loss may also occur due to factors such as increased system overheads and UE measurement deviations.
- The values of **N.S.Symbol10.NI.Avg**, **N.S.Symbol11.NI.Avg**, **N.S.Symbol12.NI.Avg**, and **N.S.Symbol13.NI.Avg** fluctuate with scheduling of UEs and may generally decrease.
- The bandwidth adaptation for interference avoidance function has the following network impacts:
 - The RRC reconfiguration probability increases, the service drop rate of the NR cell (indicated by **Service Call Drop Rate (CU, Inactive)**) increases, and the cell change success rate (indicated by **Inter-SgNB PSCell Change Success Rate (CU)** and **Intra-SgNB PSCell Change Success Rate (CU)**) may decrease.
 - Before and during BWP1 reconfiguration for a UE, the **User Uplink Average Throughput (DU)** and **User Downlink Average Throughput (DU)** decrease.
 - When the downlink interference or SRS interference is large and the uplink interference is small, the available uplink bandwidth decreases after BWP1 is reconfigured. As a result, the **User Uplink Average Throughput (DU)** may decrease.
 - The bandwidth adaptation for interference avoidance function is not recommended if the average wideband CQI in single-codeword transmission in a cell would decrease by less than 1 and the average uplink interference and noise received by each PRB (indicated by **N.UL.NI.Avg**) would increase by less than 10 dBm in LTE and NR intra-frequency networking. If this function is enabled in this case, the **User Downlink Average Throughput (DU)** may decrease.
- After the PUCCH frequency-domain position optimization function is enabled, PUCCHs are configured at one end of BWP1. This decreases the coverage by 0

dB to 1 dB when there is no strong interference and by 0 dB to 4 dB when there is strong interference.

- The measurement methods of the following external CHR counters change after the SRS SINR correction function is enabled: UL RANK1 SRS SINR0, UL Normalization Average SRS SINR, and SRS SINR. As a result, the values of these counters may change. The change varies with the SRS SINR difference between the interfered and non-interfered bands.

4.3 Requirements

4.3.1 Licenses

This function is not under license control.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RA T	Function Name	Function Switch	Reference	Description
NR TD D	CSI-RS rate matching	CSIRS_RATEMATCH_SW option of the NRDUCellPdsch.RateMatchSwitch parameter	<i>Downlink Scheduling</i>	In scenarios other than high-speed railways, the interference avoidance during PUSCH/PDSCH scheduling function requires CSI-RS rate matching to improve PDSCH symbol utilization.
NR TD D	UE bandwidth adaptation	UE_BW_ADAPTIVE_SW option of the NRDUCellBwp.BwpConfigSwitch parameter	<i>Scalable Bandwidth</i>	The bandwidth adaptation for interference avoidance function requires UE bandwidth adaptation.

RA T	Function Name	Function Switch	Reference	Description
NR TD D	Interference avoidance during PUSCH/ PDSCH scheduling	UL_CELL_LEVEL_IN TRF_AVOID and DL_UE_LEVEL_INTR F_AVOID options of the NRDUCellIntrfIdent .IntrfOptimization- Method parameter NRDUCellIntrfIdent .IntrfIdentification- Method set to UL_DL_MANUAL NRDUCellIntrfBand .InterferenceBan- dIndex set to 1 NRDUCellIntrfBand .StartRb set to 0 or NRDUCellIntrfBand .EndRb set to the total number of RBs minus 1	<i>LTE and NR Zero Bufferzone</i>	The bandwidth adaptation for interference avoidance function requires that the interference avoidance during PUSCH/PDSCH scheduling function be enabled only in scenario 2. The specific configuration requirements are as follows: • Only one interfered band can be configured (that is, the NRDUCellIntrfBand.InterferenceBandIndex parameter is set to 1). • In addition, the interfered band must be at one end of the spectrum. That is, the NRDUCellIntrfBand.StartRb parameter is set to 0, or the NRDUCellIntrfBand.EndRb parameter is set to the total number of RBs minus 1.

RA T	Function Name	Function Switch	Reference	Description
NR TD D	SRS period adaptation	SRS_PERIOD_ADAP T_SW option of the NRDUCellSrs.SrsAlg oSwitch parameter	<i>Channel Managemen t</i>	When SRS period adaptation is enabled, the NRDUCellQciBear- er.QciSpecificSrsPer iod parameter needs to be set to SL40 for all QCI so that SRSs are not transmitted across the entire bandwidth. This ensures the accuracy of SRS interference measurement.

4.3.2.2 Mutually Exclusive Functions

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentat ion)	Description
NR TD D	PDSCH Precise PMI Weight Switch	PDSCH_PRECISE_P MI_SW option of the NRDUCellAlgoSwit ch.FullChannelCovE nhSwitch parameter	None	Due to the limited aperiodic CSI-RS reporting capability, the PDSCH Precise PMI Weight Switch must be turned off when LTE and NR zero bufferzone takes effect.
NR TD D	External interference avoidance (low- frequency TDD)	UL_UE_LEVEL_INTR F_AVOID option of the NRDUCellIntrfIdent .IntrfOptimization- Method parameter and NRDUCellIntrfIdent .IntrfIdentification- Method set to UL_MANUAL or UL_AUTO	<i>Interference Avoidance</i>	Interference avoidance during PUSCH/PDSCH scheduling and external interference avoidance cannot be enabled at the same time.

RA T	Function Name	Function Switch	Reference (<i>5G RAN Feature Documentat ion</i>)	Description
NR TD D	CSI-RS interference coordination	CSIRS_INTRF_COOR D_SW option of the NRDUCellCsirs.CsiS witch parameter	<i>Massive MIMO iBeam (Low- Frequency TDD)</i>	Interference avoidance during PUSCH/PDSCH scheduling and CSI- RS interference coordination cannot be enabled at the same time.
NR TD D	Adaptive CSI-RS multi-beam adjustment	NRDUCellCsirs.Csirs CellResourceNum set to FD_RESOURCE	<i>Massive MIMO AHR (Low- Frequency TDD)</i>	When adaptive CSI- RS multi-beam adjustment and Bi- Carrier are both enabled,
NR TD D	Bi-Carrier	BI_CARRIER_SW option of the NRDUCellFeaturesS w.MultiCarrierSwit ch parameter	None	interference avoidance during PUSCH/PDSCH scheduling cannot be enabled.

4.3.2.3 Function Impacts

RA T	Function Name	Function Switch	Reference (<i>5G RAN Feature Documentatio n</i>)	Description
NR TD D	SSB period	NRDUCell.SsbPe riod	<i>NSA Mobility Management</i>	The SSB period is set to MS5 in scenarios other than high-speed railways.
NR TD D	CSI Period	NRDUCellCsirs.C siPeriod	<i>Channel Management</i>	The CSI period is set to SLOT10 in scenarios other than high-speed railways.

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR TD D	CSIRS Cell Resource Number	NRDUCELLCSIRS .CsirsCellResour- ceNum	<i>Channel Management</i>	In scenarios other than high-speed railways, when LTE and NR zero bufferzone is enabled, the number of resources for periodic CSI-RS transmission in the cell is subject to the following restrictions: • The number of resources for periodic CSI-RS transmission in the cell must be set to 1_RESOURCE when the NRDUCELLCSIRS.CsiPeriod parameter is set to SLOT20. • The number of resources for periodic CSI-RS transmission in the cell cannot be set to 4_RESOURCE when the NRDUCELLCSIRS.CsiPeriod parameter is set to SLOT40.

RA T	Function Name	Function Switch	Reference (5G RAN Feature Documentation)	Description
NR TD D	DRX	BASIC_DRX_SW option of the NRDUCellUePwr Saving.NrDuCell DrxAlgoSwitch parameter	<i>DRX</i>	LTE and NR zero bufferzone requires continuous aperiodic CSI-RS reporting. After this function is enabled, continuous aperiodic CSI-RS scheduling is performed in the uplink. In this case, the UE does not enter the DRX sleep time.
NR TD D	Low latency and high reliability	HIGH_RELIABI TY_BASIC_SW option of the NRDUCellAlgoS witch.HighRelia bilitySwitch parameter	<i>URLLC</i>	When the interfered bands conflict with the fixed resources reserved for high- reliability services, you are advised not to reserve fixed resources for high- reliability services.
NR TD D	Smart PDCCH symbol allocation	PDCCH_SYM_SM ART_ALLOC_SW option of the NRDUCellPdcch. PdcchAlgoEnhS witch parameter	<i>Massive MIMO Uplink Boosting (Low- Frequency TDD)</i>	When channel adaptation optimization in LTE and NR zero bufferzone takes effect, enabling smart PDCCH symbol allocation may reduce the increase in the success rate of uplink CCE allocation to UEs and therefore may decrease the gains in the average uplink UE throughput.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations excluding the BTS5929R. 3900 series base stations must be configured with the BBU3910.

Boards

All NR TDD-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All RF modules that work in band n38/n40/n41 support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

Cells

The cell bandwidth must be 30, 40, 50, 60, 70, 80, 90, or 100 MHz.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Interference Optimization Method	NRDUCellIntrfIdent.Int rfOptimizationMethod	Select the UL_CELL_LEVEL_INTRF_AVOID and DL_UE_LEVEL_INTRF_AVOID options and deselect other options.
Interference Identification Method	NRDUCellIntrfIdent.Int rfIdentificationMethod	Set this parameter to UL_DL_MANUAL .

Parameter Name	Parameter ID	Setting Notes
Interference Band Index	NRDUCellIntrfBand.<i>InterferenceBandIndex</i>	Set this parameter based on the frequency-domain position where interference occurs. A maximum of two bands can be configured.
Start RB	NRDUCellIntrfBand.<i>StartRb</i>	Take scenario 3 in a 100 MHz NR cell as an example. - If one 40 MHz interfered band needs to be configured, set the start RB and end RB to 164 and 272, respectively. - If two 20 MHz interfered bands need to be configured, set the start RB and end RB to 164 and 219, respectively for the first interfered band, and to 220 and 272, respectively for the second interfered band. If other cell bandwidths are used, set the parameters based on the overlapping bands between LTE and NR.
End RB	NRDUCellIntrfBand.<i>EndRb</i>	
Interference Direction	NRDUCellIntrfBand.<i>InterferenceDirection</i>	Set this parameter to UPLINK_DOWNLINK .
Channel Measurement Control Switch	NRDUCellIntrfIdent.<i>ChannelMeasCtrlSw</i>	Select the NONINTRF_SPCT_PRI_SW option to enable interference avoidance during initial access scheduling.
Interference Band 1 Spct Eff Weight Factor	NRDUCellIntrfIdent.<i>InterfBand1SpctEffWtFactor</i>	Set this parameter based on the actual interference. If the interference strength in the first interfered band is strong, set this parameter to a smaller value to reduce the scheduling probability in this interfered band. If the interference strength in the first interfered band is weak, set this parameter to a larger value to increase the scheduling probability in this interfered band.

Parameter Name	Parameter ID	Setting Notes
Interference Band 2 Spct Eff Weight Factor	NRDUCellIntrfIdent.Int rfBand2SpctEffWtFacto r	Set this parameter based on the actual interference. If the interference strength in the second interfered band is strong, set this parameter to a smaller value to reduce the scheduling probability in this interfered band. If the interference strength in the second interfered band is weak, set this parameter to a larger value to increase the scheduling probability in this interfered band.
Intrf-free Band Spct Eff Weight Factor	NRDUCellIntrfIdent.NoI ntrfBandSpctEffWtFac tor	Set this parameter based on the actual interference. A smaller value of this parameter results in a smaller impact of the non-interfered bands on full-band spectral efficiency calculation. A larger value of this parameter results in the opposite effects. If this parameter is set to a value that matches the LTE interference condition, the average cell throughput increases.
UL Interference Difference Threshold	NRDUCellIntrfIdent.U/I ntrfDiffThld	Set this parameter based on site requirements. A smaller value of this parameter results in a higher probability of triggering uplink interference avoidance. A larger value of this parameter results in a lower probability. If this parameter is set to a value that matches the uplink interference condition, the average cell throughput increases.
Common Channel Interference Avoid Switch	NRDUCellAlgoSwitch.C ommChnIntrfAvoid- Switch	Select the PUCCH_INTRF_AVOID_SW option when the PUCCH experiences interference.

Parameter Name	Parameter ID	Setting Notes
DL Interference Identification Offset	NRDUCellIntrfIdent.DIIntrfIdentOfs	Set this parameter based on site requirements. A smaller value of this parameter results in a higher probability of triggering downlink interference avoidance. A larger value of this parameter results in a lower probability. The more the match between the parameter value and the actual interference level on the live network, the higher the interfered band detection accuracy and downlink UE throughput.
Channel Measurement Control Switch	NRDUCellIntrfIdent.ChnMeasCtrlSw	Select the BW_EVAL_MCS_SEL_OPT_SW option if downlink scheduling avoidance is determined based on the SRS interference strength.
UL SRS Interference Difference Threshold	NRDUCellIntrfIdent.ULSrsIntrfDiffThld	Set this parameter based on site requirements.
Channel Measurement Control Switch	NRDUCellIntrfIdent.ChnMeasCtrlSw	Select the CHANNEL_ADAPT_OPT_SW option if the channel adaptation optimization function is required.
Channel Measurement Control Switch	NRDUCellIntrfIdent.ChnMeasCtrlSw	Select the UL_SINR_CORRECTION_SW option if the SRS SINR correction function is required.
Channel Measurement Control Switch	NRDUCellIntrfIdent.ChnMeasCtrlSw	Select the INTRF_AVOID_BW_ADAPT_SW option if the bandwidth adaptation for interference avoidance function is required.
PUCCH Algorithm Switch	NRDUCellPucch.PucchAlgoSwitch	Select the PUCCH_FREQ_POS_OPT_SW option if the PUCCH frequency-domain position optimization function is required.

NOTE

If UEs in a 100 MHz NR cell do not support measurement of multiple interfered bands and interference exists across the 40 MHz overlapping band, set the **NRDUCellIntrfBand.StartRb** and **NRDUCellIntrfBand.EndRb** parameters to **164** and **272**, respectively, for the first interfered band.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Note that changing the value of the **NRDUCellIntrfIdent.IntrfIdentificationMethod** or **NRDUCellIntrfIdent.IntrfOptimizationMethod** parameter will cause the cell to automatically restart.

Activation Command Examples

```
//Using an NR cell with a bandwidth of 100 MHz as an example
//Enabling LTE and NR zero bufferzone and adding interference configurations
MOD NRDUCELLINTRFIDENT: NrDuCellId=101,
IntrfOptimizationMethod=UL_CELL_LEVEL_INTRF_AVOID-1&DL_UE_LEVEL_INTRF_AVOID-1,IntrfIdentificationMethod=UL_DL_MANUAL;
ADD NRDUCELLINTRFBAND: NrDuCellId=101, InterferenceBandIndex=0, StartRb=164, EndRb=219,
InterferenceDirection=UPLINK_DOWNLINK;
ADD NRDUCELLINTRFBAND: NrDuCellId=101, InterferenceBandIndex=1, StartRb=220, EndRb=272,
InterferenceDirection=UPLINK_DOWNLINK;
//Enabling the interference avoidance during initial access scheduling function
MOD NRDUCELLINTRFIDENT: NrDuCellId=101,ChnMeasCtrlSw=NONINTRF_SPCT_PRI_SW-1;
//Configuring the spectral efficiency weighting coefficients and uplink interference difference threshold
MOD NRDUCELLINTRFIDENT: NrDuCellId=101, IntrfBand1SpctEffWtFactor=80,
IntrfBand2SpctEffWtFactor=80, NoIntrfBandSpctEffWtFactor=10, UlintrfDiffThld=20;
//Enabling the interference avoidance during PUCCH scheduling function
MOD NRDUCELLALGOSWITCH: NrDuCellId=101, CommChnIntrfAvoidSwitch=PUCCH_INTRF_AVOID_SW-1;
//Configuring the DL interference identification offset
MOD NRDUCELLINTRFIDENT: NrDuCellId=101, DlintrfIdentOfs=0;
//Enabling downlink scheduling avoidance determination based on the SRS interference strength
MOD NRDUCELLINTRFIDENT: NrDuCellId=101, ChnMeasCtrlSw=BW_EVAL_MCS_SEL_OPT_SW-1;
//Configuring the SRS interference difference threshold
MOD NRDUCELLINTRFIDENT: NrDuCellId=101, UlsrsIntrfDiffThld=60;
//Enabling the channel adaptation optimization function
MOD NRDUCELLINTRFIDENT: NrDuCellId=101, ChnMeasCtrlSw=CHANNEL_ADAPT_OPT_SW-1;
//Enabling the SRS SINR correction function
MOD NRDUCELLINTRFIDENT: NrDuCellId=101, ChnMeasCtrlSw=UL_SINR_CORRECTION_SW-1;
//Enabling the bandwidth adaptation for interference avoidance function
MOD NRDUCELLINTRFIDENT: NrDuCellId=101, ChnMeasCtrlSw=INTRF_AVOID_BW_ADAPT_SW-1;
//Enabling the PUCCH frequency-domain position optimization function
MOD NRDUCELLPUCCH: NrDuCellId=101, PucchAlgoSwitch=PUCCH_FREQ_POS_OPT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling LTE and NR zero bufferzone
RMV NRDUCELLINTRFBAND: NrDuCellId=101, InterferenceBandIndex=0;
```



```
RMV NRDUCELLINTRFBAND: NrDuCellId=101, InterferenceBandIndex=1;
MOD NRDUCELLINTRFIDENT: NrDuCellId=101,
IntrfOptimizationMethod=UL_CELL_LEVEL_INTRF_AVOID-0&DL_UE_LEVEL_INTRF_AVOID-0,IntrfIdentificationMethod=NOT_CONFIG;
//Disabling the interference avoidance during initial access scheduling function
MOD NRDUCELLINTRFIDENT: NrDuCellId=101,ChnMeasCtrlSw=NONINTRF_SPCT_PRI_SW-0;
//Disabling downlink scheduling avoidance determination based on the SRS interference strength
MOD NRDUCELLINTRFIDENT: NrDuCellId=101, ChnMeasCtrlSw=BW_EVAL_MCS_SEL_OPT_SW-0;
//Disabling the channel adaptation optimization function
MOD NRDUCELLINTRFIDENT: NrDuCellId=101, ChnMeasCtrlSw=CHANNEL_ADAPT_OPT_SW-0;
//Disabling the SRS SINR correction function
MOD NRDUCELLINTRFIDENT: NrDuCellId=101, ChnMeasCtrlSw=UL_SINR_CORRECTION_SW-0;
//Disabling the bandwidth adaptation for interference avoidance function
MOD NRDUCELLINTRFIDENT: NrDuCellId=101, ChnMeasCtrlSw=INTRF_AVOID_BW_ADAPT_SW-0;
//Disabling the PUCCH frequency-domain position optimization function
MOD NRDUCELLPUCCH: NrDuCellId=101, PucchAlgoSwitch=PUCCH_FREQ_POS_OPT_SW-0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

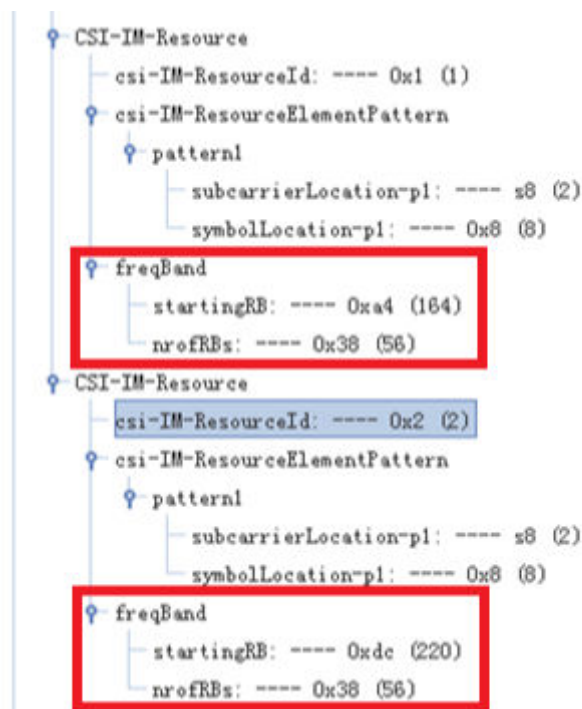
Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

- In scenarios other than high-speed railways:

After LTE and NR zero bufferzone is enabled, observe the RRC_CONN_RECFG message traced over the Uu interface to check whether the feature has taken effect. As shown in [Figure 4-6](#), the signaling in the red boxes indicates that the interfered band has been configured by the user. **startingRB** indicates the start RB position, and **nrofRBs** indicates the total number of interfered RBs.

Figure 4-6 RRC_CONN_RECFG message



The activation verification methods for the bandwidth adaptation for interference avoidance function are as follows:

- a. On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. The **Signaling Trace Management** page is displayed.
- b. Perform the following operations based on the networking mode of the NR cell:
 - If the NR cell uses SA networking, choose **Uu Interface Trace** in the navigation tree on the left. In the displayed **Uu Interface Trace** dialog box, check whether the bandwidth indicated by the *carrierBandwidth* IE in the RRCReconfiguration message is consistent with the values of the **NRDUCellBwp.UlMinCarrierBw** and **NRDUCellBwp.DlMinCarrierBw** parameters.
 - If the NR cell uses NSA networking, choose **X2 Interface Trace** in the navigation tree on the left. In the displayed **X2 Interface Trace** dialog box, check whether the bandwidth indicated by the *carrierBandwidth* IE in the SgNB Addition Request Acknowledge message is consistent with the values of the **NRDUCellBwp.UlMinCarrierBw** and **NRDUCellBwp.DlMinCarrierBw** parameters.
- c. If they are consistent, UE-level interference avoidance has taken effect. Otherwise, this function does not take effect.
- In high-speed railway scenarios, observe the values of the KPIs listed in [4.4.3 Network Monitoring](#) before and after this feature is enabled. If the values of the indicators change as expected after this feature is enabled, this feature has taken effect.

4.4.3 Network Monitoring

After this feature is enabled, observe the change in the KPIs listed in [Table 4-2](#) to check whether it has taken effect.

Table 4-2 Expected KPI results

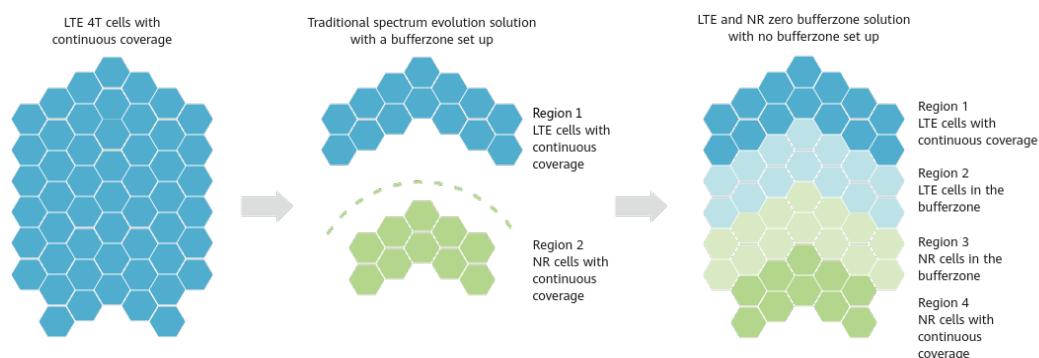
KPI	Expected Change
Cell Downlink Average Throughput (DU)	Increases
Cell Uplink Average Throughput (DU)	Increases
User Downlink Average Throughput (DU)	Increases
User Uplink Average Throughput (DU)	Increases
Downlink Resource Block Utilizing Rate (DU)	Decreases
Uplink Resource Block Utilizing Rate (DU)	Decreases

5 LTE and NR Zero Bufferzone (FDD)

5.1 Principles

As spectrum refarming from LTE to NR is time-consuming and not all LTE spectrum will be refarmed to carry NR services, some LTE bands still overlap with NR bands, resulting in intra-frequency interference between LTE and NR. Intra-frequency interference affects the traffic volume and user-perceived rate in NR cells. This is particularly evident in the initial phase of NR network construction. [Figure 5-1](#) illustrates the principles.

Figure 5-1 LTE and NR intra-frequency co-deployment



LTE and NR zero bufferzone includes the following functions:

- Automatic identification of LTE and NR bufferzones
- LNR CRS muting

Automatic Identification of LTE and NR Bufferzones

This feature supports automatic identification of LTE and NR bufferzones and configures the relationships with the identified interfering neighboring cells for the base station through MML commands for optimized interference reduction. For details, see the feature parameter description of the MAE-Optimization.

In this case, the **REFARMING_FLAG** option of the **EutraInterNFreq.AggregationAttribute** parameter and the **REFARMING_FLAG** option of the **NrNFreq.AggregationAttribute** parameter must be both selected for the target frequency for reconstruction.

Before enabling this function, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

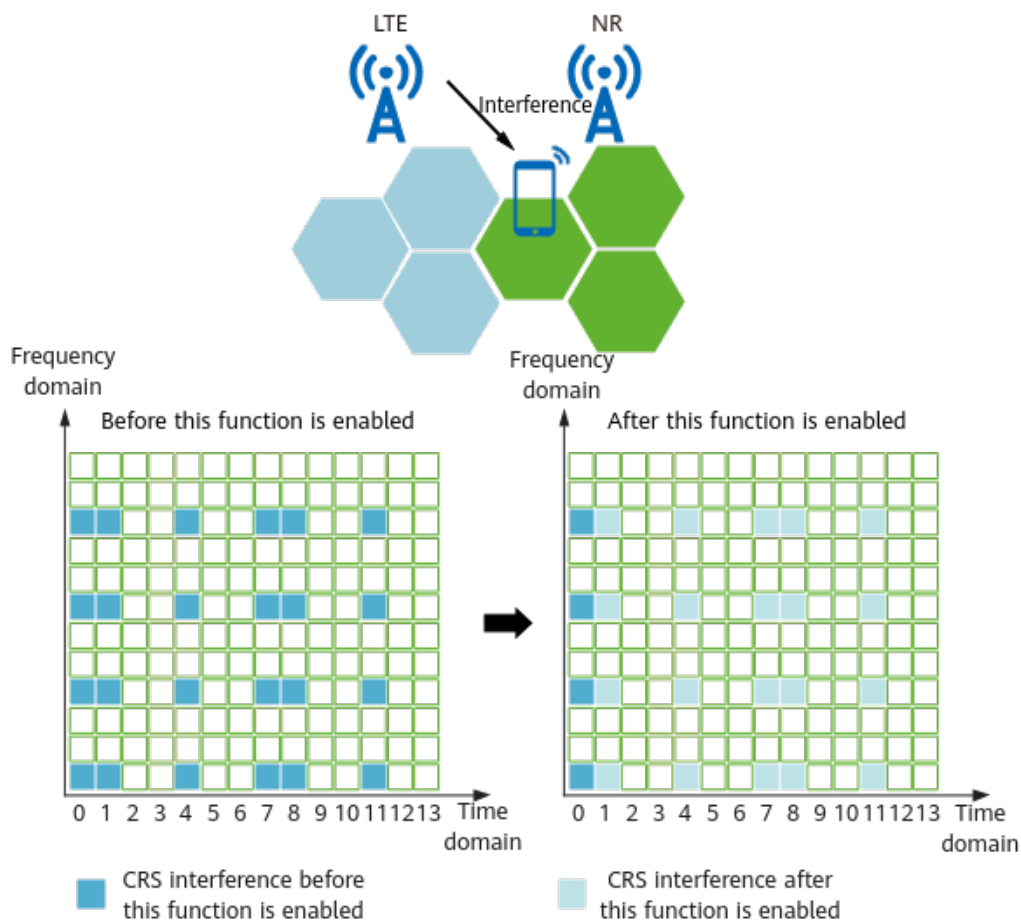
NOTICE

If automatic identification of LTE and NR bufferzones is not configured on the MAE, the **LNR_BUFFERZONE_INTRF_FLAG** option of the **NRCellEutraNRelation.LteNrIntrfIndicator** parameter must be selected to configure a maximum of 180 LTE and NR bufferzone interfering neighboring cells. In this way, this feature can be enabled.

LNR CRS Muting

If the cells served by the neighboring eNodeB are lightly loaded, LTE CRSs from the neighboring eNodeB cause strong interference to the NR cells served by the local gNodeB, as shown in [Figure 5-2](#).

Figure 5-2 LNR CRS muting



LNR CRS muting can be enabled to **mitigate the interference of LTE CRSs from the cells served by the neighboring eNodeB to the NR cells served by the local gNodeB by shutting down LTE CRS or reducing the CRS power. This improves NR performance.** Inter-base-station information exchange requires X2 links.

NOTE

The X2 interface is invoked every 5 minutes during inter-base-station negotiation over X2 links. In this case, LNR CRS muting cannot take effect immediately after it is enabled.

For details about information transmitted over the X2 link, see *X2 and S1 Self-Management in NSA Networking* and *X2 Self-Management in SA Networking*.

In SA networking, the following switches must be turned on to set up X2 links:

- LTE-NR SA X2 setup switch on the LTE side (controlled by the **LNR_SA_GNB_X2_SETUP_SW** option of the **eNodeBOamParam.InterfaceSonConfigSwitch** parameter)
- SA X2 setup switch on the NR side (controlled by the **X2_SETUP_FOR_SA_SW** option of the **gNBInterfaceParam.X2ConfigSw** parameter)

LNR CRS muting can be enabled through the following parameter settings:

- LTE side
 - Selecting the **LNR_CRS_MUTE_SW** option of the **CellAlgoExtSwitch.LnrZeroBufferzoneSw** parameter

- Setting the **CellAlgoExtSwitch.LnrCrsIntrfReduceStartTime** and **CellAlgoExtSwitch.LnrCrsIntrfReduceStopTime** parameters
- Setting the **CellAlgoExtSwitch.LteCrsPwrReduceAmount** parameter to reduce interference to NR
- Setting the **CellAlgoExtSwitch.MaxLteCrsPwrReduceDuration** parameter
- NR side
 - Selecting the **LNR_CRS_INTRF_REDUCE_SW** option of the **NRDUCellResShrCfg.LnrZeroBufferZoneSw** parameter
 - Setting the **NRDUCellResShrCfg.LnrCrsIntrfReduceStartTime** and **NRDUCellResShrCfg.LnrCrsIntrfReduceStopTime** parameters

LNR CRS muting applies to scenarios where the downlink LTE load is light.

After LTE and NR zero bufferzone is enabled, to further reduce the interference from LTE to NR, the eNodeB can adjust the On Duration periods of different UEs to increase the proportion of UEs that simultaneously enter DRX sleep state. This adjustment function takes effect when the number of UEs in a cell is less than the threshold specified by the **CellDlschAlgo.BPUeOdAlignThd** parameter. For details about DRX On Duration, see DRX and Signaling Control.

5.2 Network Analysis

5.2.1 Benefits

This feature applies to the scenarios where a small number of LTE bands yet to be refarmed in NR coverage areas overlap the NR bands and impose intra-frequency interference on NR, causing NR network performance deterioration. It is recommended that this feature be enabled only in peak rate tests in 8T8R or 32T32R scenarios. After this feature is enabled, the data rates of NR UEs performing large-packet services in the downlink in drive tests increase.

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

LNR CRS muting has the following impacts on the LTE side:

- **Cell Downlink Average Throughput, User Uplink Average Throughput, and User Downlink Average Throughput** may decrease. If dynamic DRX is enabled in a cell, dynamic DRX proactively enables UEs to enter the out-of-synchronization state. In addition, throughput fluctuation may increase the online duration of UEs, which may prolong the UE out-of-synchronization duration.
- The PRB usage of the cell increases.
- The average downlink MCS index decreases.

$$\text{Average downlink MCS index} = \frac{\sum (n \times \text{L.ChMeas.PDSCH.MCS}.n)}{\sum (\text{L.ChMeas.PDSCH.MCS}.n)}$$
 where n ranges from 0 to 28.
- The values of certain indicators may fluctuate, including downlink IBLER and RBLER.

Downlink IBLER = $(L.Traffic.DL.SCH.QPSK.ErrTB.Ibler + L.Traffic.DL.SCH.16QAM.ErrTB.Ibler + L.Traffic.DL.SCH.64QAM.ErrTB.Ibler + L.Traffic.DL.SCH.256QAM.ErrTB.Ibler) / (L.Traffic.DL.SCH.QPSK.TB + L.Traffic.DL.SCH.16QAM.TB + L.Traffic.DL.SCH.64QAM.TB + L.Traffic.DL.SCH.256QAM.TB) \times 100\%$

Downlink RBLER = $(L.Traffic.DL.SCH.QPSK.ErrTB.Rbler + L.Traffic.DL.SCH.16QAM.ErrTB.Rbler + L.Traffic.DL.SCH.64QAM.ErrTB.Rbler + L.Traffic.DL.SCH.256QAM.ErrTB.Rbler) / (L.Traffic.DL.SCH.QPSK.TB + L.Traffic.DL.SCH.16QAM.TB + L.Traffic.DL.SCH.64QAM.TB + L.Traffic.DL.SCH.256QAM.TB) \times 100\%$

- The number of OFDM symbols used by the PDCCH, the number of used CCEs, and SRB traffic may increase.
- The average RSRP or RSRQ measured by UEs in a cell decreases when LNR CRS muting is in effect and is restored to normal when this function does not take effect. Therefore, the values of the indicators related to the number of reporting times for measurement reports (such as intra-frequency neighboring cell measurement reports) and the number of resources occupied by the measurement reports may increase.
- The number of intra- and inter-frequency handovers may fluctuate.
- The number of PCC/PCell changes in CA and NSA scenarios may fluctuate. The number of SCC/SCell removals and changes may increase.
- The accuracy of RSRP grids slightly decreases.
- It results in a change in PUCCH resource allocation for CQI reporting, triggering PUCCH RB range extension at an earlier time. Consequently, it increases uplink PRB usage and may increase uplink interference. It may also lead to fluctuations in uplink access indicators (such as the random access success rate), **User Uplink Average Throughput**, uplink MCS indexes, uplink IBLERs, and uplink RBLERs.

Random access success rate = $(L.RA.GrpA.ContResolution + L.RA.GrpB.ContResolution + L.RA.Dedicate.HO.Msg3Rcv) / (L.RA.GrpA.Att + L.RA.GrpB.Att + L.RA.Dedicate.HO.Att) \times 100\%$

- The number of UEs that report CQIs at short intervals decreases and the number of UEs that report CQIs at long intervals increases.
 - If UEs that report CQIs at short intervals are all close to the cell center, the average of reported CQIs decreases.
 - If UEs that report CQIs at short intervals are all far away from the cell center, the average of reported CQIs increases.

UEs will encounter the following issues if performing RSRP and CQI measurements on the entire bandwidth when CRS transmissions are disabled:

- The measured RSRP values are inaccurate and the reported CQIs fluctuate by 0.5.
- These UEs may experience premature handovers, delayed handovers, ping-pong handovers, or ping-pong cell reselections.
- There may be slight deterioration in mobility KPIs (such as handover success rates) and accessibility KPIs (such as **RRC Setup Success Rate** and **E-RAB Setup Success Rate**). There may be slight increases in retainability KPIs (such as service drop rates), the number of random access times, the numbers of

RRC connection establishments and reestablishments, downlink residual block error rates (RBLERs), and packet loss rates.

- The preceding influences may increase slightly after the function of increasing the CRS transmission disabling rate takes effect.

5.3 Requirements

5.3.1 Licenses

RAT	Feature ID	Feature Name	Model	License Control Item	NE	Sales Unit
LTE FDD	LCOFD-2 20302	LTE and NR Zero Bufferzone (LTE FDD)	LT1SLNZ BLF00	LTE and NR Zero Bufferzone (LTE FDD)	Macro eNodeB/Lamp Site eNodeB	per Cell
NR FDD	FOFD-10 0251	LTE and NR Zero Bufferzone (NR FDD)	NR0SLNZBFZ00	LTE and NR Zero Bufferzone (NR FDD)	Macro gNodeB/Lamp Site gNodeB	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CotT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

RA T	Function Name	Function Switch	Reference	Description
NR FD D	X2 self-removal based on X2 usage	X2SON_DEL_FOR_X2USAGE_SWITCH option of the GNBX2SONCONFIG.X2SonConfigSwitch parameter	<i>X2 and S1 Self-Management in NSA Networking</i>	LNR CRS muting requires X2 self-removal based on X2 usage.
LTE FD D	X2 SON Delete Timer for LTE-NR X2 Usage	EnodebAlgoExtSwitch.X2SonDelTimerForLNX2Usage set to a value greater than 60	<i>X2 and S1 Self-Management in NSA Networking</i>	LNR CRS muting requires X2 SON Delete Timer for LTE-NR X2 Usage to be set to a value greater than 60 .
LTE FD D	X2 self-setup between LTE and NR	LTE_NR_X2_SON_SETUP_SW option of the GlobalProcSwitch.InterfaceSetupPolicySw parameter	<i>X2 and S1 Self-Management in NSA Networking</i>	LNR CRS muting requires X2 self-setup between LTE and NR.
NR FD D	X2 self-setup	X2SON_SETUP_SWITCH option of the gNBX2SonConfig.X2SonConfigSwitch parameter	<i>X2 and S1 Self-Management in NSA Networking</i>	LNR CRS muting requires X2 self-setup.
LTE FD D	LNR SA gNodeB X2 link setup	LNR_SA_GNB_X2_SETUP_SW option of the eNodeBOamParam.InterfaceSonConfigSwitch parameter	<i>X2 and S1 Self-Management in NSA Networking</i>	LNR CRS muting requires LNR SA gNodeB X2 link setup in SA networking.
NR FD D	SA X2 setup	X2_SETUP_FOR_SA_SW option of the gNBInterfaceParam.X2ConfigSw parameter	<i>X2 Self-Management in SA Networking</i>	LNR CRS muting requires X2 setup in SA networking.
LTE FD D	DRX	CellDrxPara.DrxAlgSwitch	<i>DRX and Signaling Control</i>	LNR CRS muting requires DRX.

RA T	Function Name	Function Switch	Reference	Description
LTE FDD	MR/MDT and multi-carrier unified scheduling measurement enhancement	MR_MDT_MC_SCH_MEAS_ENH_SW option of the EnodebAlgoExtSwitch.NsaProtocolCompatSw parameter	None	Automatic identification of LTE and NR bufferzones requires the MR_MDT_MC_SCH_MEAS_ENH_SW option to be selected.

5.3.2.2 Mutually Exclusive Functions

RA T	Function Name	Function Switch	Reference	Description
LTE FDD	Dynamic carrier shutdown	CARRIER_DYN_SHUTDOWN_SW option of the CellShutdownCarrierShutdownEnhancedSw parameter	<i>Energy Conservation and Emission Reduction</i>	LNR CRS muting and dynamic carrier shutdown cannot be both enabled.
LTE FDD	CRS dynamic muting	CRS_DYNAMIC_MUTING_SW option of the SymbolPwrSaving.SymbolShutdownEnhancedSw parameter	<i>Energy Conservation and Emission Reduction</i>	LNR CRS muting and CRS dynamic muting cannot be both enabled.
LTE FDD	Channel isolation for PIM interference avoidance	PIM_INTERF_CHN_SHUTDOWN_SW option of the PimInterfSmartAvoid.PimAlgoSwitch parameter	<i>Interference Detection and Suppression</i>	LNR CRS muting and channel isolation for PIM interference avoidance cannot be both enabled.
LTE FDD	LTE CRS overhead optimization	LTE_CRN_OVERHEAD_OPT_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>	LNR CRS muting and LTE CRS overhead optimization cannot be both enabled.
LTE FDD	CSPC	CellCspcPara.CellCspcSwitch	<i>CSPC</i>	LTE CRS power reduction and CSPC cannot be both enabled.

RA T	Function Name	Function Switch	Reference	Description
LTE FD D	eCSPC	CellCspcPara.CelleC spcSwitch	<i>CSPC</i>	LTE CRS power reduction and eCSPC cannot be both enabled.
LTE FD D	Intra- eNodeB CSPC	CellCspcPara.IntraE nbCspcSw	<i>CSPC</i>	LTE CRS power reduction and intra-eNodeB CSPC cannot be both enabled.
LTE FD D	Adaptive SFN/SDMA	CellAlgoSwitch.Sfn DISchSwitch	<i>SFN</i>	LTE CRS Power Reduction Amount can be set to a value other than 0 only when joint scheduling is used in SFN/SDMA scenarios.
LTE FD D	Inter- eNodeB SFN based on eNodeB coordination	Number of CellAuxEnbBind MOs ≥ 1	<i>SFN</i>	LTE CRS Power Reduction Amount can be set to a value other than 0 only when no CellAuxEnbBind MO is configured for the cell.
LTE FD D	Inter- eNodeB adaptive SFN/SDMA based on eNodeB coordination	Number of CellAuxEnbBind MOs ≥ 1	<i>SFN</i>	LTE CRS Power Reduction Amount can be set to a value other than 0 only when no CellAuxEnbBind MO is configured for the cell.
LTE FD D	In-band relay	InBandRelayDeNbS witch option of the CellAlgoSwitch.Rel aySwitch parameter	<i>Relay</i>	LTE CRS power reduction and in-band relay cannot be both enabled.
LTE FD D	ePDCCH	EpdccchFunctionS- witch option of the CellPdcchAlgo.Epdcc chAlgoSwitch parameter	<i>Physical Channel Resource Managemen t</i>	LTE CRS power reduction and ePDCCH cannot be both enabled.

RA T	Function Name	Function Switch	Reference	Description
NB- IoT	NB-IoT cell	Cell.NbCellFlag	<i>NB-IoT Basics (FDD)</i>	LTE CRS power reduction is not supported in NB-IoT cells.
LTE FD D	Short TTI	SHORT_TTI_SW option of the CellShortTtiAlgo.St tiAlgoSwitch parameter	<i>Short TTI (FDD)</i>	LTE CRS power reduction and short TTI cannot be both enabled.
LTE FD D	CRS power reduction	PDSCHCFG.CrsPwrR eductionAmount	None	If LTE CRS Power Reduction Amount is set to a value other than 0 , PDSCHCFG.CrsPwrR eductionAmount must be set to DB_0 .
LTE FD D	PHICH inner-loop power control	PhichInnerLoopPcS witch option of the CellAlgoSwitch.DIP cAlgoSwitch parameter	<i>Power Control</i>	LTE CRS power reduction and PHICH inner-loop power control cannot be both enabled.
LTE FD D	Massive MIMO assistance cell	Cell.MultiRruCellM ode set to ASSISTED_MASSIVE _MIMO_CELL	None	LTE CRS power reduction is not supported in massive MIMO assistance cells.
LTE FD D	MPRU aggregation cell	Cell.MultiRruCellM ode set to MPRU_AGGREGATI ON	None	LTE CRS power reduction is not supported in MPRU aggregation cells.
LTE FD D	Digital combination cell	Cell.MultiRruCellM ode set to DIGITAL_COMBINA TION	None	LTE CRS power reduction is not supported in digital combination cells.
LTE FD D	RB reservation in a cell	CellRbReserve MO configured for a cell, with CellRbReserve.RbRs vType set to DOWNLINK_MODE	None	LTE CRS power reduction and RB reservation in a cell cannot be both enabled.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	OTDOA-based positioning	OtdoaSwitch option of the ENodeBAlgoSwitch . LcsSwitch parameter or the NB_OTDOA_SWITCH option of the CellLcsCfg . CellLcsSwitch parameter	<i>LCS</i>	LTE CRS power reduction and OTDOA-based positioning cannot be both enabled.
LTE FDD	eMBMS	CellMBMSCfg . MBMSSwitch	<i>eMBMS</i>	LTE CRS power reduction and eMBMS cannot be both enabled.
LTE FDD	Breathing Pilot	BreathingPilotSwitch option of the CellDlschAlgo . BreathingPilotAlgoSwitch parameter	<i>Breathing Pilot</i>	LNR CRS muting and Breathing Pilot cannot be both enabled.
LTE FDD	Coordinated pilot scheduling	EmimoCpsSwitch option of the CellAlgoSwitch . EmimoSwitch parameter	<i>eMIMO (FDD)</i>	LNR CRS muting and coordinated pilot scheduling cannot be both enabled.

5.3.2.3 Function Impacts

- Device-related functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Low power consumption mode	CellLowPower . LowPwrSwitch	<i>Green Site</i>	LNR CRS muting does not take effect in a cell that has entered low power consumption mode.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	SOC level-based low power consumption mode	• IPWRM.S OCLVLAD JSW • SOCLVL_B ASSED_LO W_PWR_SW option of the CellLowPower.Low PwrConsumeModeExtSw parameter	<i>Green Site</i>	LNR CRS muting does not take effect in a cell that has entered SOC level-based low power consumption mode.
LTE FDD	RRU PA efficiency improvement	RE.PAEFFSWITCH	<i>RRU PA Efficiency Improvement</i>	LNR CRS muting provides lower gains in cells where RRU PA efficiency improvement is enabled.

- NSA networking functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	NSA networking based on EPC	NSA_DC_CAPABILITY_SWITCH option of the NsaDcMgmt Config.NsaDcAlgoSwitch parameter	<i>EPC-based NSA Basics</i>	After LNR CRS muting is enabled, the data volume of an NSA UE may be lower than the downlink data volume threshold for large-packet services and consequently this UE is considered as a non-large-packet UE. As a result, the gains brought by LNR CRS muting are affected.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	SN-terminated MCG bearer transmission	SN_TERM_N_MCG_BEARER_SWITCH option of the NsaDcMgmtConfig.NsaDcAlgoSwitch parameter	<i>EPC-based NSA Performance Enhancement</i>	After LNR CRS muting is enabled, the downlink data volume of NSA UEs with SN-terminated MCG bearers may be lower than the downlink data volume threshold for large-packet services and consequently such UEs are identified as non-large-packet UEs. As a result, the gains brought by LNR CRS muting are affected.

- Basic functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Preallocation	PreAllocationSwitch option of the CellAlgoSwitch.UISwitch parameter	<i>Uplink Scheduling</i>	As normal preallocation increases the chances of uplink scheduling, the CRS transmission disabling rate will decrease after LNR CRS muting is enabled.
LTE FDD	Smart preallocation	SmartPreAllocationSwitch option of the CellAlgoSwitch.UISwitch parameter	<i>Uplink Scheduling</i>	If smart preallocation is used in a cell, it is difficult for UEs to enter the DRX sleep state and LNR CRS muting can hardly be performed in the cell.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	PDCCH aggregation level adaptation	CellPdcchAlgo.AggLvlSelStrageForDualCW	<i>Physical Channel Resource Management</i>	LNR CRS muting increases the dual-codeword reporting probability. To reduce the impact of PDCCH DTX on network performance in dual-codeword scenarios, it is recommended that the CellPdcchAlgo.AggLvlSelStrageForDualCW parameter be set to STRATEGYBASEDON COVERAGE .
LTE FDD	DL 256QAM	DL256QamSwitch option of the CellAlgoSwitch.DL256QamAlgoSwitch parameter	<i>Modulation Schemes</i>	Inter-cell interference decreases, and the proportion of times DL 256QAM is selected increases.
LTE FDD	Adaptive power allocation for cell-edge UEs	AdptCellEdgePwrAllocSw option of the CellAlgoSwitch.CellDlCoverEnhanceSwitch parameter	<i>Downlink Scheduling</i>	Adaptive power allocation for cell-edge UEs is not performed in the TTIs where LNR CRS muting is in effect.

- Interference mitigation functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	SFN cell	None	<i>SFN</i>	In SFN networking, interference between intra-frequency cells has been reduced. In this scenario, the gains offered by LNR CRS muting will decrease.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Network-assisted CRS interference cancellation, or network-assisted interference cancellation and suppression	CellAlgoSwitch.CrsIcSwitch/ CellAlgoSwitch.NaicsSwitch	<i>Network Assisted Interference Cancellation</i>	If there are UEs that support neighboring cell CRS interference cancellation, the interference may have already been mitigated. In this situation, the User Downlink Average Throughput may decrease after LNR CRS muting is enabled.
LTE FDD	Intelligent PIM interference avoidance	PIM_INTERF_SMART_AVOID_SWITCH option of the PimInterfSmartAvoid.PimAlgoSwitch parameter	<i>Interference Detection and Suppression</i>	LNR CRS muting does not take effect at the PIM interference detection phase of intelligent PIM interference avoidance. LNR CRS muting can take effect at the PIM interference avoidance phase of intelligent PIM interference avoidance.
LTE FDD	Super combined cell	None	<i>Super Combined Cell (FDD)</i>	LNR CRS muting offers lower gains in such cells because intra-frequency interference is already low in these cells.

- CoMP functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	DL CoMP with TM10	FDDHomNetDLCompSwitch and FDDHetNetDLCompSwitch options of the CellAlgoSwitch.DLCompSwitch parameter	<i>DL CoMP (FDD)</i>	LNR CRS muting does not take effect for TM10 JT UEs.
LTE FDD	DL CoMP with TM9	Tm9JtSwitch option of the CellAlgoSwitch.DLCompSwitch parameter	<i>DL CoMP (FDD)</i>	LNR CRS muting does not take effect for TM9 JT UEs.

- MIMO functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Data-learning-based downlink intelligent AMC	DL_INTEL_AMC_SELECTION_SW option of the CellIntelAmcConfig.IntelligentAmcSwitch parameter	<i>eMIMO (FDD)</i>	LNR CRS muting affects the sample collection for downlink intelligent AMC. Therefore, the model-based prediction effect of downlink intelligent AMC may deteriorate, which may affect the gains provided by downlink intelligent AMC.
LTE FDD	Downlink MU-MIMO in TM9	4TxTM9MuMimoSwitch option of the CellAlgoSwitch.EmimoSwitch parameter	<i>eMIMO (FDD)</i>	Downlink MU-MIMO in TM9 does not take effect when LNR CRS muting is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Downlink MU-MIMO in TM4	EmimoMuMimoSwitch option of the CellAlgoSwitch . EmimoSwitch parameter	<i>eMIMO (FDD)</i>	Downlink MU-MIMO in TM4 does not take effect when LNR CRS muting is in effect.
LTE FDD	WTTx MU-MIMO	Tm9FourLayerPairSwitch option of the CellAlgoExtSwitch . WttxMuMimoSwitch parameter	<i>WTTx MU-MIMO (FDD)</i>	WTTx MU-MIMO does not take effect when LNR CRS muting is in effect.
LTE FDD	Downlink intelligent JAMC	CELL_DL_JOINT_INTEL_AMC_SW option of the CellIntelAmcConfig . IntelligentAmcSwitch parameter	<i>Smart 8T8R (FDD)</i>	LNR CRS muting affects the sample collection for downlink intelligent JAMC. Therefore, the model-based prediction effect of downlink intelligent JAMC may deteriorate, which may affect the gains provided by downlink intelligent JAMC.
LTE FDD	Downlink intelligent joint power control	DL_INTEL_JOINT_PC_SW option of the CellDISchExtParam . DISchExtSwitch parameter	<i>Smart 8T8R (FDD)</i>	LNR CRS muting does not take effect in the TTIs where power reduction in downlink intelligent joint power control is in effect.
LTE FDD	Downlink intelligent joint AMC and power control	DL_INTEL_JOINT_AMC_AND_PC_SW option of the CellDISchExtParam . DISchExtSwitch parameter	<i>Smart 8T8R (FDD)</i>	LNR CRS muting does not take effect in the TTIs where power reduction in downlink intelligent joint AMC and power control is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Power increase in downlink intelligent joint power control	CellDlSchExtParam.DlPcCrspwrIncreaseAmt	<i>Smart 8T8R (FDD)</i>	Power increase in downlink intelligent joint power control does not take effect in subframes where LNR CRS muting is in effect.
LTE FDD	Enhanced extreme power sharing	LTE_ENH_EXT_POWER_SHARING_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>Smart 8T8R (FDD)</i>	If a cell shares PAs with other cells, it cannot use the shared power of those cells in TTIs where LNR CRS muting is in effect in the cell.
LTE FDD	Centralized power sharing	LTE_CENTRAL_POWER_SHARING_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>Smart 8T8R (FDD)</i>	If a cell shares PAs with other cells, it cannot use the shared power of those cells in TTIs where LNR CRS muting is in effect in the cell.
LTE FDD	Hybrid power sharing	LTE_HYBRID_POWER_SHARING_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>Smart 8T8R (FDD)</i>	If a cell shares PAs with other cells, it cannot use the shared power of those cells in TTIs where LNR CRS muting is in effect in the cell.
LTE FDD	Massive MIMO downlink coordinated power control	MM_DL_COORD_PC_SW option of the SectorSplitGroup.IntelligentBeamSwitch parameter	<i>Massive MIMO Enhancements (FDD)</i>	LNR CRS muting does not take effect in the TTIs where power reduction is performed in massive MIMO downlink coordinated power control.

- Power management functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Dynamic Power Sharing Between LTE Carriers	LTE_DYN_POWER_SHARING_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>Dynamic Power Sharing Between LTE Carriers</i>	If a cell shares PAs with other cells, it cannot use the shared power of those cells in TTIs where LNR CRS muting is in effect in the cell.
LTE FDD	Extreme power sharing	LTE_EXTREME_POWER_SHARING_SW option of the CellDynPowerSharing.DynamicPowerSharingSwitch parameter	<i>Dynamic Power Sharing Between LTE Carriers</i>	If a cell shares PAs with other cells, it cannot use the shared power of those cells in TTIs where LNR CRS muting is in effect in the cell.
LTE FDD	PDSCH power boosting	PDSCHCfgo.PdschPowerBoostingSwitch	<i>Power Control</i>	PDSCH power boosting does not take effect in the TTIs where LNR CRS muting is in effect.

- Energy saving functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Intelligent power-off of carriers in the same coverage	CellShutdown.CellShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	LNR CRS muting offers lower gains in cells where the "intelligent power-off of carriers in the same coverage" function is enabled.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Multi-carrier coordinated energy saving	CellShutdown.CellShutdownSwitch set to ON_MULTI_CARRIER_HIER_SHUTDOWN or ON_MULTICARRIER_HIERMANUAL_SHUTDOWN	<i>Energy Conservation and Emission Reduction</i>	LNR CRS muting does not take effect when multi-carrier coordinated energy saving is in effect.
LTE FDD	Dynamic carrier shutdown phase 2	CellPwrSavingAlgo.CarrierDynShutdownPh2Switch set to CAPACITY_CELL	<i>Energy Conservation and Emission Reduction</i>	LNR CRS muting does not take effect for a cell in the dynamic carrier shutdown state.
LTE FDD	Automatic cell shutdown	CellAutoShutdown.CellAutoShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	LNR CRS muting does not take effect when automatic cell shutdown is in effect.
LTE FDD	Intelligent power-off of carriers in the same coverage as UMTS networks	InterRatCellShutdown.ForceShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	LNR CRS muting does not take effect when intelligent power-off of carriers in the same coverage as UMTS networks is in effect.
LTE FDD	RF channel intelligent shutdown	CellRfShutdown.RfShutdownSwitch	<i>Energy Conservation and Emission Reduction</i>	LNR CRS muting does not take effect when RF channel intelligent shutdown is in effect.
LTE FDD	RF module regular time sleep	eNodeBAutoPowerOff.AutoPowerOffSwitch	<i>Energy Conservation and Emission Reduction</i>	LNR CRS muting does not take effect when RF module regular time sleep is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Dynamic voltage adjustment	CellAlgoSwitch.DynAdjVoltageSwitch	<i>Energy Conservation and Emission Reduction</i>	When the triggering conditions for dynamic voltage adjustment and LNR CRS muting are both met, LNR CRS muting will take effect but dynamic voltage adjustment will not.
LTE FDD	Enhanced symbol power saving	MBSFNShutDownSwitch option of the CellAlgoSwitch.DlSchSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When enhanced symbol power saving and LNR CRS muting are both enabled, enhanced symbol power saving preferentially takes effect. LNR CRS muting does not take effect when enhanced symbol power saving is in effect.
LTE FDD	Enhanced RF channel dynamic muting	RF_CHN_DYNAMIC_MUTING_ENH_SW option of the CellRfChnDynamicMuting.RfChannelDynMutingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	If both LNR CRS muting and RF channel dynamic muting are enabled, the L.ChMeas.DFEE.RFChannelMutingTti.Number and L.Cell.PilotOffTti.Number counter values will fluctuate. These two counters indicate the number of TTIs where RF channel dynamic muting takes effect and the number of TTIs where some CRSs are not transmitted, respectively.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Flexible power allocation in RF channel dynamic muting	RF_DYN_MUTE_FLEX_POWER_ALLOC_SWITCH option of the CellRfChnDynamicMuting.RfChannelDynamicMutingAlgoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	Flexible power allocation in RF channel dynamic muting does not take effect when LNR CRS muting is in effect.
LTE FDD	8T beam mode shift for energy saving	BEAM_MODE_SHIFT_SWITCH option of the SectorSplitGroup.SplitCellGroupOptionsSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	LNR CRS muting does not take effect during the shift between multi-beam mode and single-beam mode in 8T beam mode shift for energy saving.
LTE FDD	32T beam mode shift for energy saving	BEAM_MODE_SHIFT_SWITCH option of the SectorSplitGroup.SplitCellGroupOptionsSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	LNR CRS muting does not take effect during the shift between the four-beam and two-beam modes in 32T beam mode shift for energy saving.
LTE FDD	Basic symbol power saving	SymbolShutdownSwitch option of the ENodeBAlgoSwitch.PowerSaveSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	LNR CRS muting offers lower gains in cells where basic symbol power saving is enabled.
LTE FDD	RF module deep dormancy	RRU.DORMANCY_SWITCH set to ON or ON_DEEP_SLEEP	<i>Multi-RAT Coordinated Energy Saving</i>	When LNR CRS muting and RF module deep dormancy are both enabled, RF module deep dormancy preferentially takes effect. LNR CRS muting does not take effect when RF module deep dormancy is in effect.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	RF module super dormancy	RRU.DORMANCYSW set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	When LNR CRS muting and RF module super dormancy are both enabled, RF module super dormancy preferentially takes effect. LNR CRS muting does not take effect when RF module super dormancy is in effect.
LTE FDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATSW (LTE eNodeB, 5G gNodeB) parameter	<i>Multi-RAT Coordinated Energy Saving</i>	RF module fast deep dormancy takes precedence over LNR CRS muting when both functions are enabled. LNR CRS muting does not take effect when RF module fast deep dormancy is in effect.
LTE FDD	DRX	CellDrxParam.DrxAlgSwitch	<i>DRX and Signaling Control</i>	DRX increases the probability of LNR CRS muting taking effect and the gains brought by LNR CRS muting. If DRX parameters are set for semi-persistent scheduling, LNR CRS muting does not take effect. For details about DRX parameter configuration for semi-persistent scheduling, see <i>DRX and Signaling Control</i>.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Dynamic DRX	CellAlgoSwitch.DynDrxSwitch	<i>DRX and Signaling Control</i>	The DRX status of UEs in dynamic DRX mode will change after LNR CRS muting is enabled. Consequently, the L.Traffic.User.Unsyn.Drx.Avg counter value may also change. This counter indicates the average number of UEs configured with DRX parameters for the out-of-synchronization state in a cell.

- Inter-RAT coordination functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	LTE FDD and NR Flash Dynamic Spectrum Sharing	SpectrumCloud.SpectrumCloudSwitch set to LTE_NR_SPECTRUM_SHR	<i>LTE FDD and NR Dynamic Spectrum Sharing</i>	In LTE FDD and NR Flash Dynamic Spectrum Sharing scenarios, LNR CRS muting takes effect in a lower proportion of cases and consequently offers fewer benefits than in NR-only scenarios.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Hybrid DSS Based on Asymmetric Bandwidth	SpectrumCloud.SpectrumCloudSwitch set to LTE_NR_SPECTRUM_SHR and the LNR_SPECTRUM_SHR_ASYM_SW option of the SpectrumCloud.SpectrumCloudEnhSwitch parameter selected	<i>LTE FDD and NR Hybrid Dynamic Spectrum Sharing</i>	In Hybrid DSS Based on Asymmetric Bandwidth scenarios, LNR CRS muting takes effect in a lower proportion of cases and consequently offers fewer benefits than in NR-only scenarios.

- Other functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Load simulation	Number of CellSimuLoad MOs ≥ 1	None	LNR CRS muting does not take effect when one or more CellSimuLoad MOs are configured.
LTE FDD	Service guarantee for UEs running unlimited services	UsUeGuaranteeSwitch option of the CellUSParaCfg.UsAlgoSwitch parameter	None	LNR CRS muting does not take effect when there are UEs running unlimited services in a cell. If the cell is in the intelligent power control energy saving state, the cell exits LNR CRS muting.
LTE FDD	TTI Bundling	TtiBundlingSwitch option of the CellAlgoSwitch.ULSchSwitch parameter	<i>VoLTE</i>	LNR CRS muting offers lower gains in TTI bundling mode because UEs in this mode cannot enter the DRX mode.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Cell-level PDSCH power adjustment	CELL_LEVEL_PDSCH_POWER_ADJUST_W option of the CellPwrSavingAlgo.PwrSavingAlgoSwitch parameter	None	If both LNR CRS muting and cell-level PDSCH power adjustment are enabled, the power decrease specified by the CellPwrSavingPolicy.PowerAdjustmentLevel and CellPwrSavingAlgo.IntelPcCrsPwrRedAmt parameters cannot exceed 6 dB.
LTE FDD	eMTC introduction	EMTC_SWITCH option of the CellEmtcAlgo.EmtcAlgoSwitch parameter	<i>eMTC</i>	LNR CRS muting can take effect in invalid subframes in FDD cells with eMTC enabled. Therefore, at least one invalid subframe must be configured (by using the CellEmtcAlgo.EmtcDlInvalidSfCfg parameter) for LNR CRS muting to take effect in the FDD cell with eMTC enabled. A larger number of invalid subframes results in a better energy saving effect but a greater impact on eMTC. It is recommended that one to three invalid subframes be configured or that the number of invalid subframes be determined based on the eMTC load.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	Downlink coordinated power control	DL_COORD_PC_SW option of the CellMimoParaCfg.MimoSwitch parameter	<i>Low-Band Booster</i>	LNR CRS muting does not take effect in the TTIs where downlink coordinated power control is in effect for the purpose of power reduction.
LTE FDD	Neighboring frequency addition based on X2 setup messages and eNodeB configuration update messages	ADD_FREQ_CFG_SW option of the GlobalProcSwitch.X2BasedUptNcellCfgSwitch parameter	<i>ANR Management</i>	When the ADD_FREQ_CFG_SW option of the GlobalProcSwitch.X2BasedUptNcellCfgSwitch parameter is selected, enabling LNR CRS muting may affect measurements of the neighboring frequencies that are automatically added. It is recommended that the ADD_FREQ_CFG_SW option of the GlobalProcSwitch.X2BasedUptNcellCfgSwitch parameter be deselected for the neighboring base station of the base station serving the cell where LNR CRS muting is enabled.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD	CA cell	None	<i>Carrier Aggregation</i>	After LNR CRS muting takes effect, SCell RSRP measurements by CA UEs will be affected, leading to a slight change in the number of CA UEs in a cell. After LNR CRS muting takes effect, data allocation between the PCell and SCells of a CA UE will be affected and the downlink PRB usage of the PCell and SCells may fluctuate. In inter-eNodeB CA in relaxed backhaul scenarios, the increase in the average downlink UE throughput may be lowered slightly.

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations excluding the BTS5929R. 3900 series base stations must be configured with the BBU3910.

Boards

- All LTE FDD-capable and NR FDD-capable main control boards support this function.
- LTE FDD baseband processing units: UBBPe and later baseband processing units support this function.
- NR FDD baseband processing units: UBBPg and later baseband processing units support this function.

RF Modules

All LTE FDD-capable and NR FDD-capable RF modules support this function.

5.3.4 Others

This function takes effect only when both LCOFD-220302 LTE and NR Zero Bufferzone (LTE FDD) and FOFD-100251 LTE and NR Zero Bufferzone (NR FDD) are activated.

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

Table 5-1 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 5-1 LTE parameters used for activation

Parameter Name	Parameter ID	Setting Notes
LTE and NR Zero Bufferzone Switch	CellAlgoExtSwitch.LnrZeroBufferzoneSw	Select the LNR_CRS_MUTE_SW option.
LNR CRS Interference Reduce Start Time	CellAlgoExtSwitch.LnrCrsIntrfReduceStartTime	Set this parameter based on the network plan. The value of the CellAlgoExtSwitch.LnrCrsIntrfReduceStartTime parameter must be less than or equal to that of the CellAlgoExtSwitch.LnrCrsIntrfReduceStopTime parameter.
LNR CRS Interference Reduce Stop Time	CellAlgoExtSwitch.LnrCrsIntrfReduceStopTime	Set this parameter based on the network plan. The value of the CellAlgoExtSwitch.LnrCrsIntrfReduceStartTime parameter must be less than or equal to that of the CellAlgoExtSwitch.LnrCrsIntrfReduceStopTime parameter.
LTE CRS Power Reduction Amount	CellAlgoExtSwitch.LteCrsPwrReduceAmount	You are advised to use the recommended value.

Parameter Name	Parameter ID	Setting Notes
Maximum LTE CRS Power Reduction Duration	CellAlgoExtSwitch.MaxLteCrsPwrReduceDuration	You are advised to use the recommended value.
Aggregation Attribute	EutranInterNFreq.AggregationAttribute	Select the REFARMING_FLAG option of this parameter based on the network plan.
Aggregation Attribute	NrNFreq.AggregationAttribute	Select the REFARMING_FLAG option of this parameter based on the network plan.
UE DRX OD Align Threshold	CellDlschAlgo.BPUeOdAlignThd	You are advised to use the recommended value.

Table 5-2 NR parameters used for activation

Parameter Name	Parameter ID	Setting Notes
LTE and NR Zero Bufferzone Switch	NRDUCellResShrCfg.LnrZeroBufferZoneSw	Select the LNR_CRS_INTRF_REDUCE_SW option.
LNR CRS Interference Reduce Start Time	NRDUCellResShrCfg.LnrCrsIntrfReduceStartTime	Set this parameter based on the network plan. The value of the NRDUCellResShrCfg.LnrCrsIntrfReduceStartTime parameter must be less than or equal to that of the NRDUCellResShrCfg.LnrCrsIntrfReduceStopTime parameter.
LNR CRS Interference Reduce Stop Time	NRDUCellResShrCfg.LnrCrsIntrfReduceStopTime	Set this parameter based on the network plan. The value of the NRDUCellResShrCfg.LnrCrsIntrfReduceStartTime parameter must be less than or equal to that of the NRDUCellResShrCfg.LnrCrsIntrfReduceStopTime parameter.

Parameter Name	Parameter ID	Setting Notes
LTE and NR Interference Indicator	NRCCellEutraNRRelation.LteNrIntrfIndicator	Set this parameter based on the network plan. - If automatic identification of LTE and NR bufferzones is not configured on the MAE, manually select the LNR_BUFFERZONE_INTRF_FLAG option of this parameter. - If automatic identification of LTE and NR bufferzones is configured on the MAE, the LNR_BUFFERZONE_INTRF_FLAG option of this parameter is automatically selected through script delivery.

5.4.1.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- Activation command examples on the LTE side

```
//Enabling automatic identification of LTE and NR bufferzones and setting the information of the frequency to be reconstructed for identifying interfering cells
MOD EUTRANINTERNFREQ: LocalCellId=0, DlEarfcn=100, AggregationAttribute=REFARMING_FLAG-1;
MOD NRNFREQ: LocalCellId=0, DlArfcn=422890, AggregationAttribute=REFARMING_FLAG-1;
//Enabling LNR CRS muting
MOD CELLALGOEXTSWITCH: LocalCellId=0, MaxLteCrsPwrReduceDuration=20,
LteCrsPwrReduceAmount=3, LnrZeroBufferzoneSw=LNR_CRS_MUTE_SW-1,
LnrCrsIntrfReduceStartTime=00&00&00, LnrCrsIntrfReduceStopTime=06&00&00;
//Configuring UE DRX OD Align Threshold
MOD CELLDLSCHALGO: LocalCellId=0, BPUEOdAlignThd=15;
```
- Activation command examples on the NR side

```
//Enabling LNR CRS muting
MOD NRDUCELLRESSHRCFG: NrDuCellId=0, LnrZeroBufferZoneSw=LNR_CRS_INTRF_REDUCE_SW-1,
LnrCrsIntrfReduceStartTime=00&00&00, LnrCrsIntrfReduceStopTime=06&00&00;
```

If automatic identification of LTE and NR bufferzones is not configured on the MAE, configure **LNR Bufferzone Interference Flag**.

```
//(Optional) Configuring LNR Bufferzone Interference Flag only when automatic identification of LTE  
and NR bufferzones is not configured on the MAE  
MOD NRCELLEUTRANRELATION: NrCellId=0, Mcc="302", Mnc="220", EnodebId=0, CellId=0,  
LteNrIntrfIndicator=LNR_BUFFERZONE_INTRF_FLAG-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Deactivation command examples on the LTE side

```
//Disabling LNR CRS muting  
MOD CELLALGOEXTSWITCH: LocalCellId=0, LteCrspwrReduceAmount=0,  
LnrZeroBufferzoneSw=LNR_CRS_MUTE_SW-0;
```

- Deactivation command examples on the NR side

```
//Disabling LNR CRS muting  
MOD NRDUCELLRESSHRCFG: NrDuCellId=0, LnrZeroBufferZoneSw=LNR_CRS_INTRF_REDUCE_SW-0;  
//Disabling LNR Bufferzone Interference Flag  
MOD NRCELLEUTRANRELATION: NrCellId=0, Mcc="302", Mnc="220", EnodebId=0, CellId=0,  
LteNrIntrfIndicator=LNR_BUFFERZONE_INTRF_FLAG-0;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

Observe the values of the following indicators before and after this function is enabled. If the values change as expected after this function is enabled, this function has taken effect.

- The value of the **L.DLPwr.Avg** counter decreases.
- The value of the **L.DLPwr.Max** counter decreases.
- The average downlink MCS index on the NR side increases.

Average downlink MCS index on the NR side = $\sum(n \times N.ChMeas.PDSCH.MCS.n) / \sum(N.ChMeas.PDSCH.MCS.n)$, where n ranges from 0 to 28.

5.4.3 Network Monitoring

After this feature is enabled, monitor the network impacts of this feature using the performance indicators listed in [5.2.2 Impacts](#).

6 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

- Step 2** On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.
- End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

9 Reference Documents

- 3GPP TS 38.104: "NR;Base Station (BS) radio transmission and reception"
- *Technical Specifications of BBUs in 3900 & 5900 Series Base Station Product Documentation*
- Feature parameter description documents in *5G RAN Feature Documentation*:
 - *Interference Avoidance*
 - *Downlink Scheduling*
 - *NSA Mobility Management*
 - *Channel Management*
 - *DRX*
 - *URLLC*
 - *Scalable Bandwidth*
 - *Massive MIMO Uplink Boosting (Low-Frequency TDD)*
 - *Massive MIMO iBeam (Low-Frequency TDD)*
 - *Massive MIMO AHR (Low-Frequency TDD)*
- Feature parameter description documents in *eRAN Feature Documentation*:
 - *Energy Conservation and Emission Reduction*
 - *Interference Detection and Suppression*
 - *CSPC*
 - *SFN*
 - *Relay*
 - *Physical Channel Resource Management*
 - *eMTC*
 - *NB-IoT Basics (FDD)*
 - *Short TTI (FDD)*
 - *Power Control*
 - *LCS*
 - *eMBMS*
 - *Breathing Pilot*
 - *eMIMO (FDD)*

- *Green Site*
- *RRU PA Efficiency Improvement*
- *Uplink Scheduling*
- *Modulation Schemes*
- *Downlink Scheduling*
- *Network Assisted Interference Cancellation*
- *Super Combined Cell (FDD)*
- *DL CoMP (FDD)*
- *WTTx MU-MIMO (FDD)*
- *Smart 8T8R (FDD)*
- *Massive MIMO Enhancements (FDD)*
- *Dynamic Power Sharing Between LTE Carriers*
- *Power Control*
- *DRX and Signaling Control*
- *VoLTE*
- *eMTC*
- *Low-Band Booster*
- *ANR Management*
- *Carrier Aggregation*
- *X2 and S1 Self-Management in NSA Networking*
- *X2 Self-Management in SA Networking*
- *LTE FDD and NR Dynamic Spectrum Sharing*
- *LTE FDD and NR Hybrid Dynamic Spectrum Sharing*
- *EPC-based NSA Basics*
- *EPC-based NSA Performance Enhancement*
- *Multi-RAT Coordinated Energy Saving*